DATA PRODUCT CASE STUDY · STELLARALGO

PRIME: one governed data layer for a multi-tenant product

From week-old, disputed numbers to one layer every team, dashboard and AI reads from.

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Data and Business Analyst interview · The City of Calgary

One story, your six questions

S

Situation

Q1 · The problem and why it mattered

T

Task

Q1 · Who needed it, and what I owned

A

Action

Q2 · The product

Q3 · The approach

Q4 · The challenges

R

Result

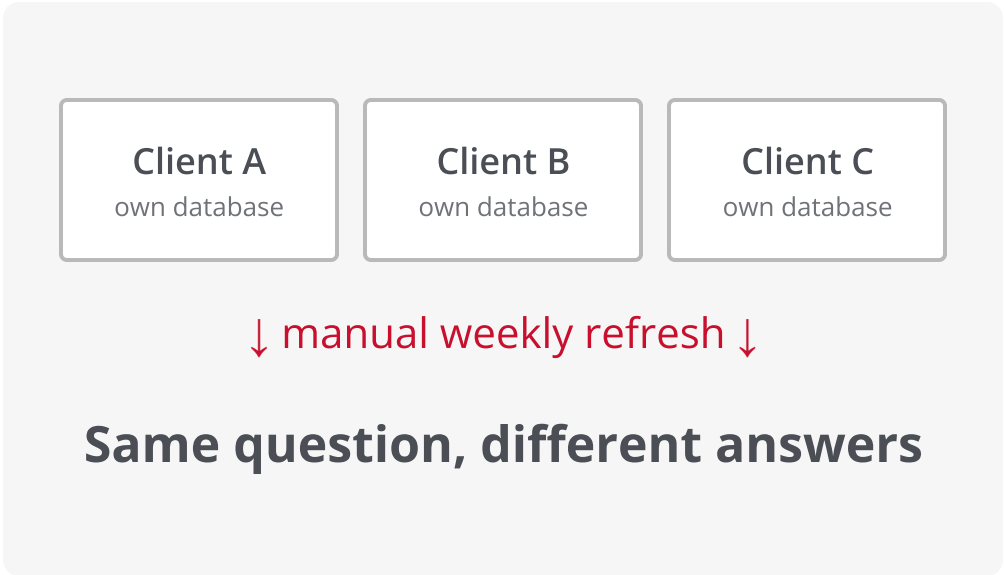
Q2 · The value

Q5 · What I'm proud of

Q6 · What I'd improve

The letters in each slide's corner show where we are.

Every client lived in its own database



Up to 7 days old

A key refresh ran weekly, by hand.

One system per client

A question across clients meant visiting each one in turn.

No clear owner

Ownership was split across teams, with no agreed service levels.

a

Why it mattered

Teams and clients worked from different numbers, and internal reporting couldn't be trusted.

One set of numbers everyone could trust

MY ROLE

Director, Product Innovation & Analytics. I owned PRIME's product management from planning to general availability.

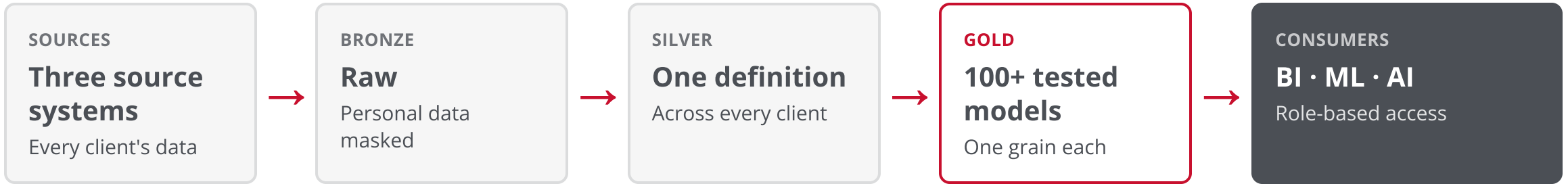
b Who needed it

Customer Success · Product · Clients · Analysts · Data Engineering

WHAT SUCCESS HAD TO MEAN

Trusted	One definition per metric, across every client.
Fresh	Daily, not weekly.
Owned	Named owners, targets and incident tiers.
Safe	Personal data masked, access by role.
Self-serve	Routine questions without an engineering ticket.

The product: one governed layer



a What it does

Transforms every client's data once, in dbt, into one Snowflake layer that any approved team or tool can use.

b How people use it

- BI dashboards
- Client data sharing
- Customer Success bot

Ask in plain English, get the dashboard's number



Each semantic model has one grain, so the AI can't double count. We shipped it first as an internal Customer Success bot. I led discovery for Stellar Insights, the client-facing version.

Four decisions that shaped the build

1 Strategy

Fix the data once, not every report. Agree the contract before the charts.

2 Plan

Bronze, silver, gold. One grain per gold table. Mask personal data on landing.

3 Build

Airflow and dbt replaced the manual weekly refresh with a daily run.

4 Review

Freshness checked against each client's target, with AI digests of pipeline errors.

a Sourcing and preparing the data

Three source systems, mapped client by client into one dimensional model, then tested.

b Methods and tools

Snowflake · dbt · Airflow · AWS · Medallion layers · Data contract · LLMs

What got in the way, and how I handled it

Type	Challenge	How I solved it
Technical	Every client system had its own shape.	One dimensional model all clients map into.
Analytical	The same metric meant different things to different teams.	One definition in silver, one documented grain per gold table.
People	Nobody owned freshness or accuracy.	A data contract I wrote: owners, targets per domain, three incident tiers.

What it delivered

✓ FRESH

~~7 days~~

< 12 hrs

Data freshness

✓ TRUSTED

100+

Tested models, replacing a database per client

✓ OWNED

30 min

To flag a missed refresh, before a client notices

✓ SELF-SERVE

No ticket

For routine Customer Success questions

✓ **SAFE** Personal data masked at the raw layer; every person and tool sees only what their role allows.

Customer Success, Product and clients now work from the same numbers, on a schedule they can rely on.

What I'm most proud of

Governance stopped being overhead and became the reason people could self-serve.

The same tested layer feeds dashboards, client data and the AI, with no separate pipeline for each. That's why the AI's answers match the dashboards.

What I'd improve with more time

Measure from day one

Baseline escalations and Customer Success hours before launch, so the value is a number.

Insights for clients

Take the internal bot to clients as Stellar Insights, on the same layer.

More semantic models

Cover more domains, each with one grain, so the AI can answer more.

The same work at the City: find where numbers disagree, agree one definition with the owners, and make it self-serve.

Thank you. Questions?

Case study codychandler.ca/work/stellaralgo-prime